CivicSim

Demographically Grounded LLM Simulations of Public Opinion for Inclusive Policy Testing.

PROJECT Master of Information Management and Systems (MIMS) — Final Capstone Report

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LIVE DEMO civicsim.xyz · CODE github.com/Sushanti99/CivicSim

DATE May 2026

Ground it before you simulate it.

Abstract

SUMMARY

Public policy is routinely shipped without a credible read on how affected populations will respond. Town halls, commissioned surveys, and stakeholder roundtables are slow, expensive, and structurally biased toward the constituents who are easiest to convene. **CivicSim** is a web-based policy-testing tool that generates demographically grounded LLM focus groups in seconds: a policymaker enters a draft policy and a location, and CivicSim returns a synthetic electorate sampled from U.S. Census microdata, conditioned on Pew American Trends Panel (ATP) opinion priors, with per-agent rationales, an aggregate sentiment distribution, and an interactive simulation log.

This report presents three diagnostic studies — each measuring a directional, structural distortion in current LLM-based opinion simulation — and the three-step framework that corrects them: (1) draw agents from Census microdata, not from the survey corpus; (2) select conditioning variables by information-gain ablation, not by marginal rank; (3) apply tiered, domain-aware geographic conditioning. The framework is operationalized as a deployed product at **civicsim.xyz**, evaluated against held-out ATP subgroup distributions and compared against vanilla LLM baselines. The methodological argument is simple: when an LLM simulates a population, the choice of *who* is being simulated is an empirical question, not a design preference.

Contents

1 Executive Summary

WHAT CIVICSIM IS

CivicSim is a policy-testing tool that generates demographically grounded LLM-based focus groups for proposed policies. A policymaker enters a draft policy and a location; CivicSim returns a synthetic electorate of Census-sampled agents, each one carrying an empirical opinion prior derived from the Pew American Trends Panel, with per-agent rationales and an aggregate sentiment distribution. The system is built around three methodological commitments that distinguish it from generic "persona prompting":

COMMIT 01

Census-grounded sampling

Agents are sampled from ACS microdata, not the survey corpus, so joint demographic distributions reflect the real population.

COMMIT 02

Information-gain selection

Conditioning variables are chosen by joint information gain over 1,426 opinion items — not by intuition or marginal rank.

COMMIT 03

Domain-aware geography

Geographic conditioning is tiered by policy domain and raised for young agents, where regional variance is up to 70% higher.

COMMIT 04

Auditable per-agent traces

Every simulation persists the sampled persona, retrieved prior, demographic backoff path, and raw LLM response for replay.

38,449
ATP RESPONDENTS

2.5M
CENSUS RECORDS

94%
DEMOGRAPHIC COVERAGE

2.4×
SIGNAL VS. BASELINE

The pilot is deployed at **civicsim.xyz**. A typical simulation of 25 agents at the regional or county level completes in under twenty seconds, costs a fraction of a cent per run, and produces a downloadable audit trail. The framework is built to be portable: every measurement we report can be re-derived from public data, and the grounding pipeline plugs into any off-the-shelf LLM.

> *Ground it before you simulate it. When an LLM simulates a population, the choice of who is being simulated is an empirical question, not a design preference.*

2 Problem & Motivation

WHY THIS MATTERS

Policymakers ship policy under three constraints that traditional public engagement tools were never built to satisfy. *Time*: a school board weighing competing funding formulas cannot wait six weeks for a survey before a budget vote. *Cost*: a county supervisor iterating on a rent stabilization ordinance cannot commission a new focus group for each variant. *Coverage*: town halls and roundtables are structurally biased toward constituents who are easy to convene — homeowners with flexible schedules, English-speakers, people with transportation, people with childcare — and systematically under-represent the populations a policy is most likely to land hardest on. The cost of these constraints is foreseeable and well documented: policies that look equitable in aggregate routinely produce sharply unequal outcomes on the ground, and those outcomes typically become visible only after rollout, when the political and human cost of changing course is highest.

Recent work on generative agents (Park et al., 2023; Stanford HAI, 2024) has shown that LLM agents seeded with sufficiently rich human data can replicate individual policy attitudes with up to 85% of human test–retest accuracy. The opportunity, in other words, is real: if the data grounding is good enough, large language models can act as a low-latency, low-cost *pre-engagement* layer — a simulation harness that lets policymakers stress-test policy variants and messaging frames against realistic constituent populations before public consultation, not in place of it.

But this opportunity is also fragile. Off-the-shelf LLMs are known to encode systematic opinion biases (Santurkar et al., 2023), and the most common grounding strategies — persona prompts, survey-only conditioning, marginal-rank variable selection — quietly fail in exactly the subpopulations a policymaker most needs to hear from. CivicSim's operational research question is whether demographically grounded generative agents, conditioned on Census microdata and seeded with real opinion data, can surface policy-relevant stakeholder concerns accurately enough to help a policymaker anticipate constituent response *before* they propose. The three empirical studies in Section 5 each measure one of those quiet failures and motivate one step of CivicSim's corrective framework.

2.1 Target Users

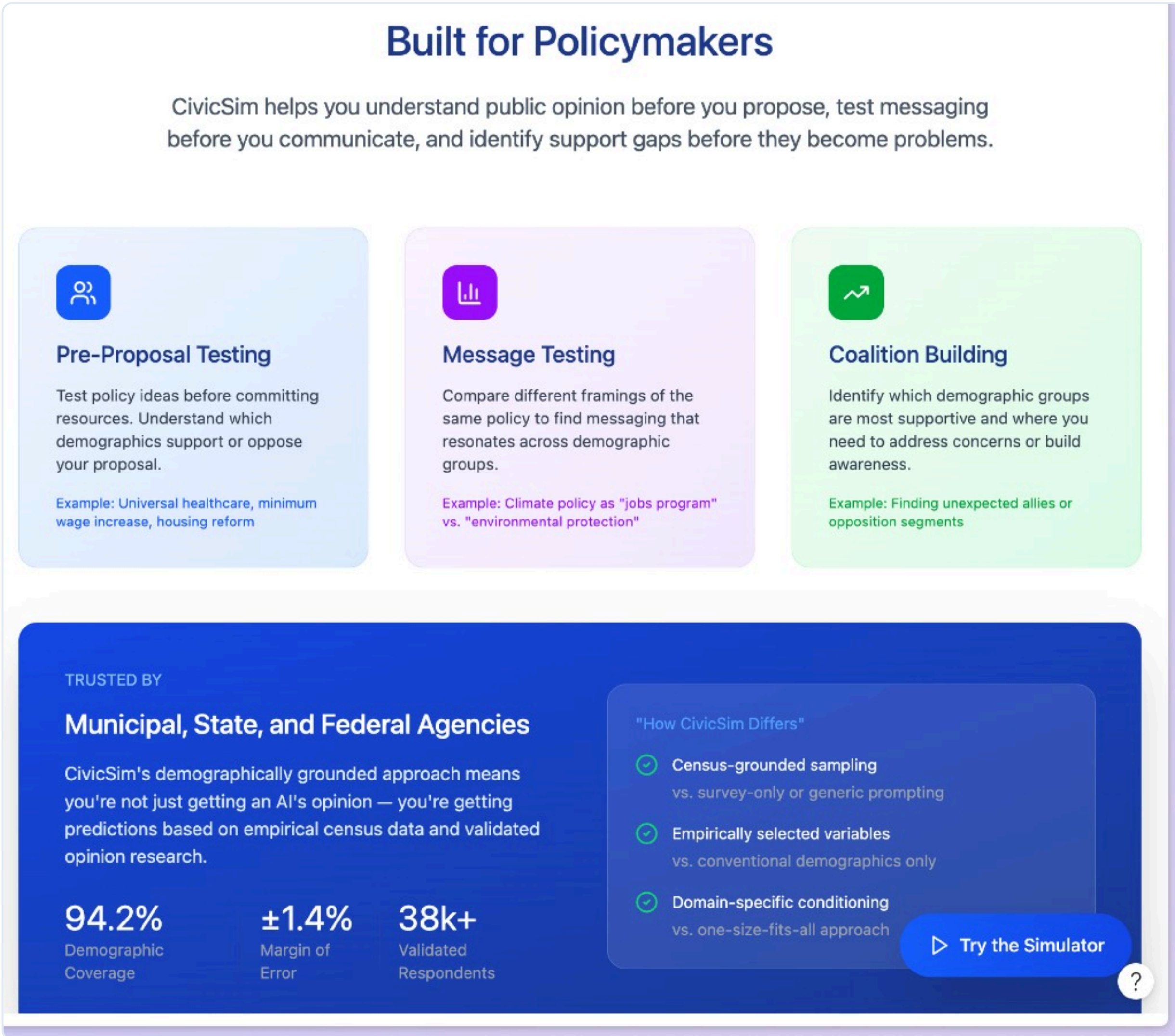


Figure 1. Three primary user workflows surfaced by CivicSim: pre-proposal testing, message-frame comparison, and coalition mapping. Source: CivicSim landing page.

CivicSim is built for three concrete workflows: **pre-proposal testing** (which demographic segments support or oppose a draft policy?), **message testing** (does framing climate policy as a "jobs program" outperform "environmental protection" in a target cohort?), and **coalition building** (where are the unexpected allies and where are the at-risk segments that need targeted outreach?). The two-group focus group output is built around those workflows: one group representing the local population, one representing the directly affected stakeholders, with sentiment distributions and per-agent rationales placed side by side for contrast.

3 Related Work

WHERE CIVICSIM SITS

CivicSim builds on three converging research literatures. The first is *generative agent simulation*: Park et al. (2023, 2024) showed that LLM agents seeded with structured interview data can reproduce individual survey responses at up to 85% of human test–retest reliability. The second is *subpopulation alignment*: the SubPOP line of work and PolyPersona (Dash et al., 2025) examined whether persona-conditioned LLMs can recover real subgroup opinion distributions, and how the choice of persona representation affects downstream alignment. The third is *opinion bias measurement*: Santurkar et al. (2023) documented systematic alignment between off-the-shelf LLMs and specific demographic cohorts (younger, more educated, more liberal), motivating any pipeline that uses LLMs as opinion proxies to ground them explicitly.

Two strands of statistical practice also feed directly into the system. Jiang et al. (2024) demonstrated a geographically explicit synthetic population generated from ACS-style joint distributions — the same methodological move CivicSim adopts at Stage 2 of its pipeline. And the long-standing literature on post-stratification weighting in survey research (Pew Research Center methodology series; Lohr, *Sampling: Design and Analysis*) provides the baseline against which our Study 1 finding — that marginal weighting fails to recover joint distributions — should be read.

CivicSim's contribution is not a new agent architecture or a new alignment benchmark. It is a deployable system that operationalizes three measurable structural distortions in current practice as three concrete pipeline choices, and ships the resulting tool as a usable product. The framework is designed to be inherited: each measurement is reproducible from public data, each pipeline choice is documented, and each per-agent response is logged and auditable.

4 System Architecture

TWO STREAMS, ONE PERSONA

CivicSim's central architectural commitment is that *who a person is* and *what a person believes* should be grounded independently before they are fused into a single conditioning persona. Most LLM persona pipelines collapse these two questions into a single prompt — "you are a 35-year-old voter who…" — which leaves both the demographic distribution and the opinion distribution entirely at the mercy of the underlying model's training-set prior. CivicSim instead operates two orthogonal grounding streams.

STRUCTURAL STREAM

Who is represented

Draws agents from ACS Public Use Microdata Sample joint distributions of age, race, income, occupation, and geography — so every demographic combination reflects an empirically observed cell of the U.S. population.

BEHAVIORAL STREAM

What they tend to believe

Seeds each agent with a Pew ATP–derived opinion prior `P(y | d, q)` matched to its demographic cell, with graceful backoff to lower-dimensional cells when the joint cell is sparse.

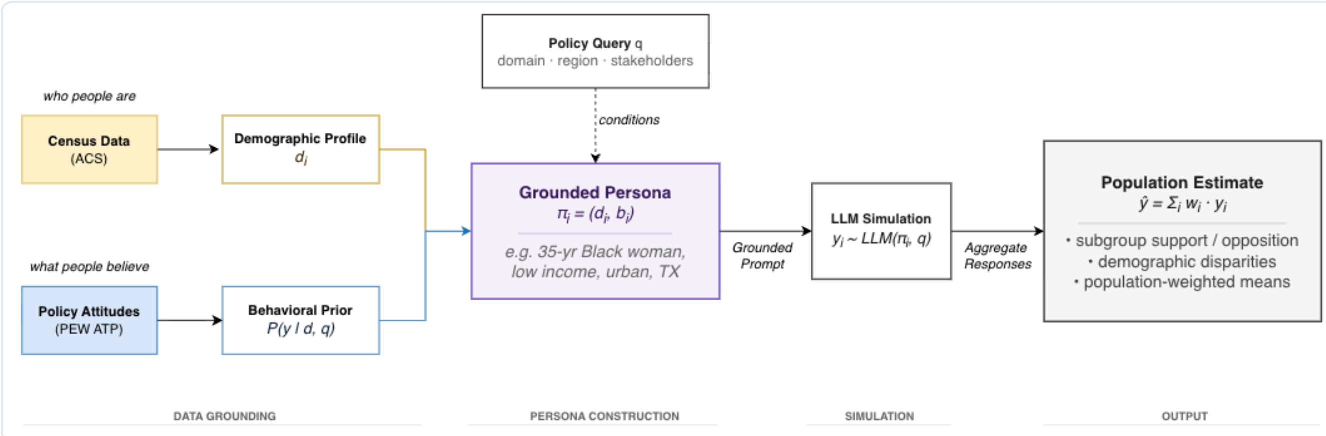


Figure 2. CivicSim's two-stream architecture. ACS PUMS yields a demographic profile d_i; Pew ATP yields a behavioral prior $P(y \mid d, q)$ conditioned on the policy query q. The two are fused into a grounded persona $\pi_i = (d_i, b_i)$, which conditions every LLM call. Aggregated responses produce subgroup support/opposition, demographic disparities, and population-weighted means.

4.1 The four-stage pipeline

Every simulation runs as a four-stage directed acyclic graph. The stages are designed so that each one can be replaced or audited independently; the system never collapses the grounding decisions into a single monolithic prompt.

STAGE 01 · INGEST

Policy & domain resolution

The user submits a natural-language policy and a domain (economy, housing, healthcare, environment, etc.), which activates a domain-specific conditioning-variable vector.

STAGE 02 · SAMPLE

Structural draw from ACS

For each agent slot, CivicSim samples a joint demographic profile from ACS PUMS — joint, not marginal, so rare-but-real combinations appear at the right rate.

STAGE 03 · MATCH

Behavioral prior retrieval

The agent's demographic vector is projected onto the ATP cell vocabulary; the nearest matching cell's answer distribution is retrieved, with graceful backoff if the cell is sparse.

STAGE 04 · SIMULATE

Grounded LLM response

The persona, retrieved prior, and policy text are assembled into a prompt; the LLM returns a stance + one-sentence rationale, which is aggregated and streamed to the UI.

The aggregator produces population-weighted sentiment, demographic-cell disparity decompositions, and a per-agent trace log. Every per-agent record persists the full prompt, the matched ATP cell, the demographic-backoff path, the model name, and the raw LLM response, so any output is replayable and externally auditable.

4.2 Demographic-cell backoff

The ATP corpus is large but not infinite: at the intersection of low income × non-metro × specific Census division × specific age bracket, many cells contain too few respondents to estimate a stable distribution. CivicSim handles this with explicit, ordered backoff: the system always attempts the most specific cell first, then drops the least-informative dimension until a populated cell is found. Geographic conditioning (Census region and Census division) is dropped *last*, not first, because the empirical study in Section 5.2 shows that geography carries the single largest unique contribution to opinion signal — discarding it early would amount to optimizing the wrong loss.

```
Backoff order (drop least-informative first):
  urbanicity  →  income_group  →  education_group  →  race_eth
            →  gender  →  age_group  →  F_CDIVISION  →  F_CREGION
```

Every backoff step is recorded in the per-agent trace, so the policymaker can see — for the specific agents that informed a specific simulation — exactly which dimensions had to be relaxed to retrieve a populated prior.

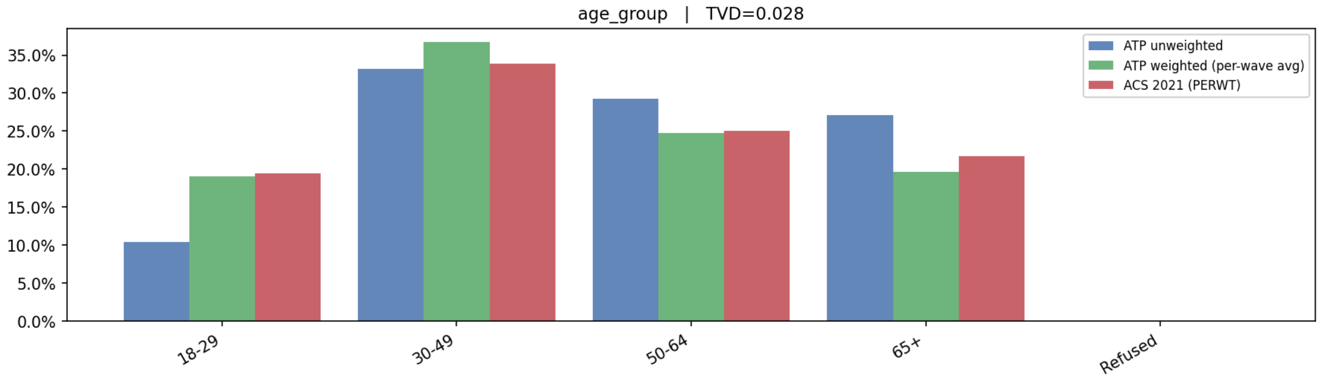
5 Empirical Findings

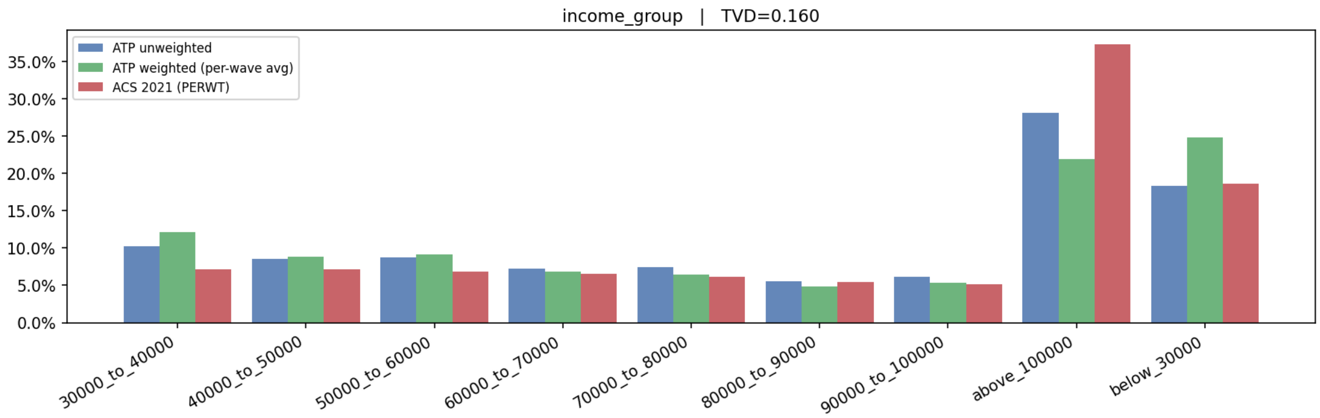
THREE FAILURES, THREE CORRECTIONS

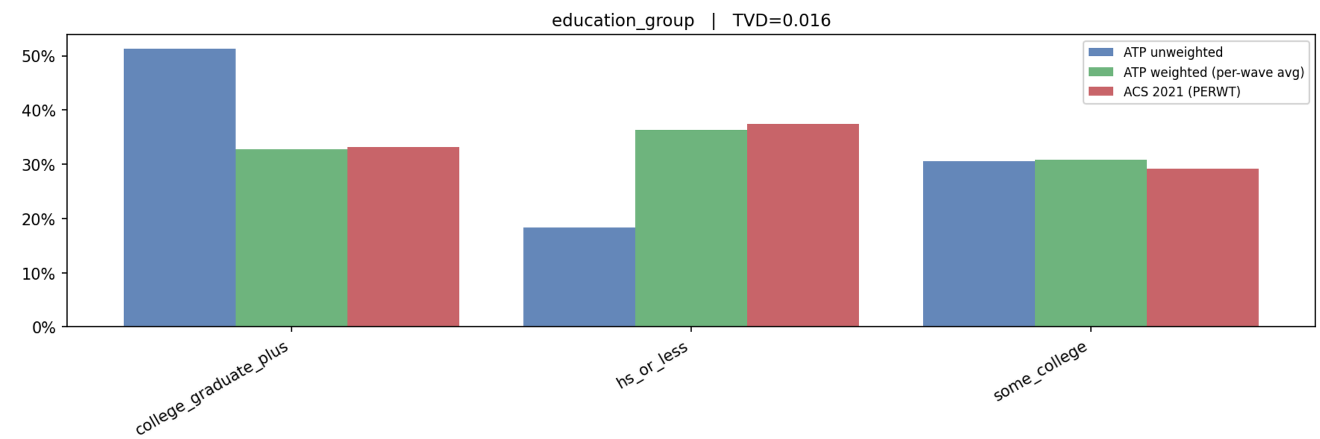
Each of the three diagnostic studies below isolates a structural distortion in current LLM-based opinion simulation and motivates exactly one step of CivicSim's framework.

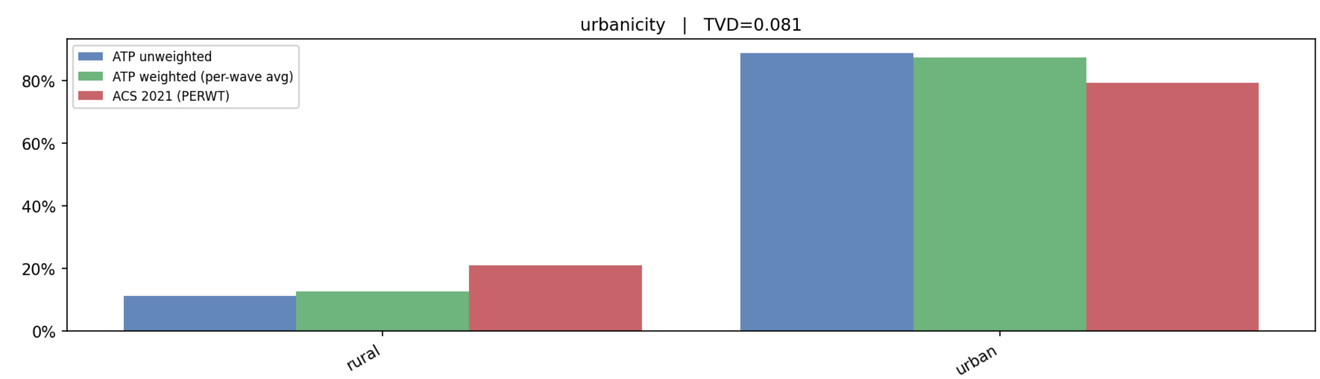
5.1 Study 1 — The survey is not the population

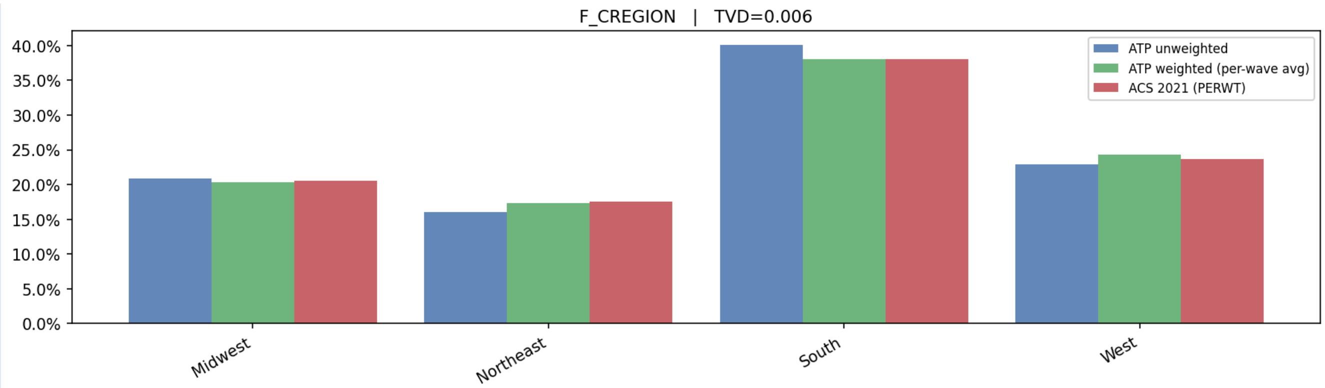
Pew ATP ships with post-stratification weights designed to correct for sampling deviations from the U.S. population. We tested whether those weights are sufficient at the *joint* distribution level: can weighted ATP substitute for ACS as the population from which agents are drawn? After weighting, marginal Total Variation Distances against the ACS are small for most dimensions — but two carry substantial residual gaps, and joint distributions inside sparse intersectional cells show misalignment an order of magnitude larger than the full-sample baseline.

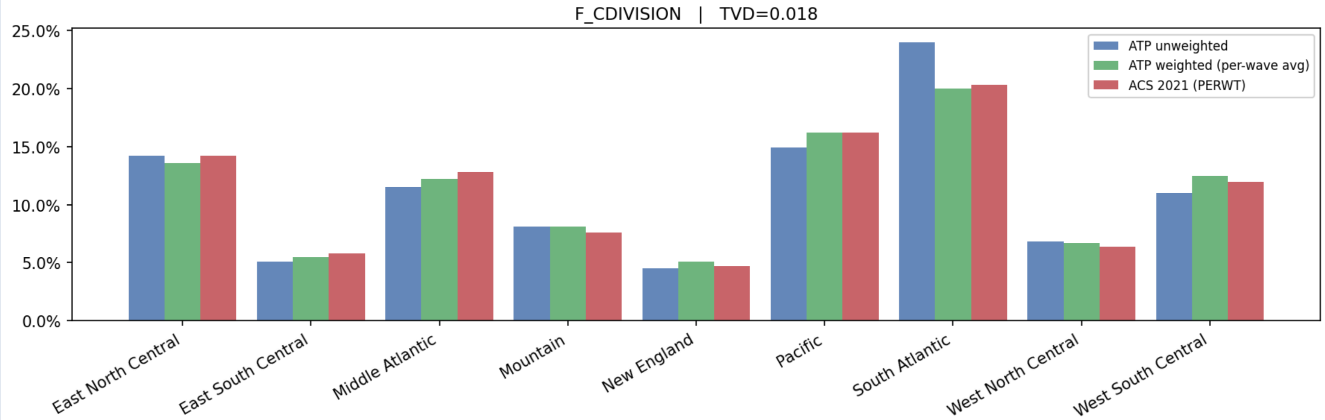

age_group | TVD=0.028
ATP unweighted
ATP weighted (per-wave avg)
ACS 2021 (PERWT)
35.0%
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
18-29
30-49
50-64
65+
Refused

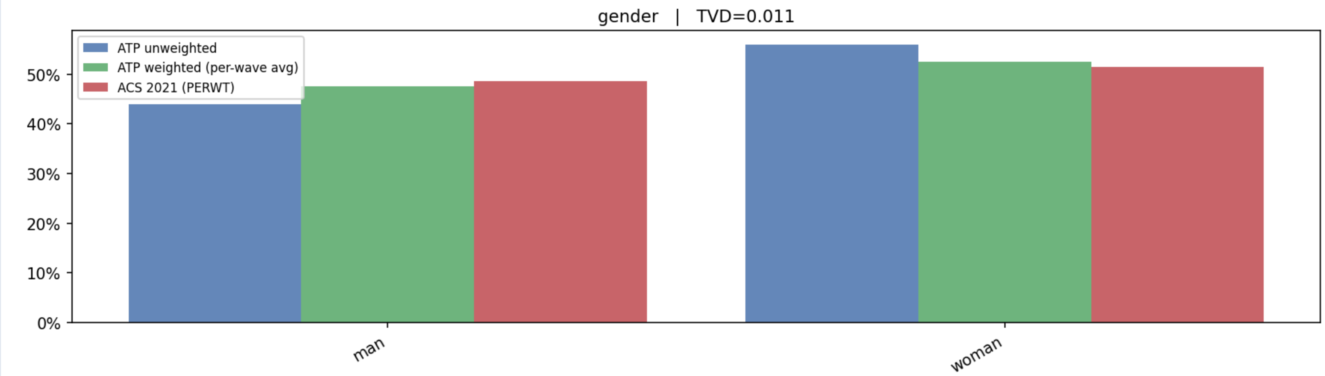

income_group | TVD=0.160
ATP unweighted
ATP weighted (per-wave avg)
ACS 2021 (PERWT)
35.0%
30.0%
25.0%
20.0%
15.0%
10.0%
5.0%
0.0%
30000_to_40000
40000_to_50000
50000_to_60000
60000_to_70000
70000_to_80000
80000_to_90000
90000_to_100000
above_100000
below_30000


education_group | TVD=0.016
ATP unweighted
ATP weighted (per-wave avg)
ACS 2021 (PERWT)
50%
40%
30%
20%
10%
0%
college_graduate_plus
hs_or_less
some_college


urbanicity | TVD=0.081
ATP unweighted
ATP weighted (per-wave avg)
ACS 2021 (PERWT)
80%
60%
40%
20%
0%
rural
urban


F_CREGION | TVD=0.006
ATP unweighted
ATP weighted (per-wave avg)
ACS 2021 (PERWT)
0.0%
5.0%
10.0%
15.0%
20.0%
25.0%
30.0%
35.0%
40.0%
Midwest
Northeast
South
West


F_CDIVISION | TVD=0.018
ATP unweighted
ATP weighted (per-wave avg)
ACS 2021 (PERWT)
0.0%
5.0%
10.0%
15.0%
20.0%
25.0%
East North Central
East South Central
Middle Atlantic
Mountain
New England
Pacific
South Atlantic
West North Central
West South Central


gender | TVD=0.011
ATP unweighted
ATP weighted (per-wave avg)
ACS 2021 (PERWT)
0%
10%
20%
30%
40%
50%
man
woman

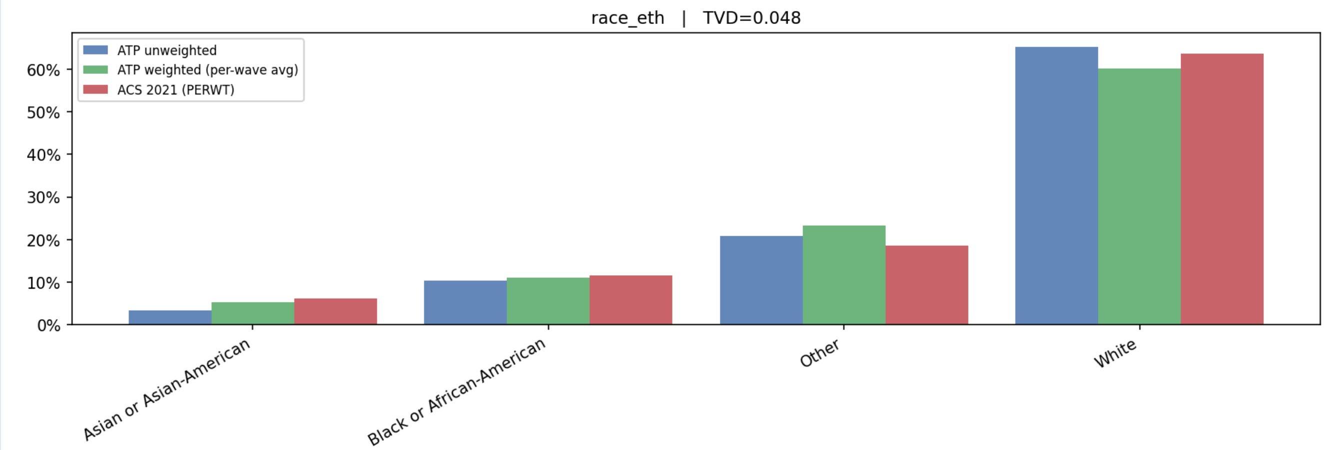

race_eth | TVD=0.048
ATP unweighted
ATP weighted (per-wave avg)
ACS 2021 (PERWT)
0%
10%
20%
30%
40%
50%
60%
Asian or Asian-American
Black or African-American
Other
White

Figure 3. ACS (true U.S. population, red) vs. weighted Pew ATP (green) and unweighted ATP (blue) across eight demographic dimensions. Marginal post-stratification weighting closes most one-dimensional gaps (TVD ≤ 0.05 for age, region, gender, education, race), but cannot recover joint distributions in systematically excluded subgroups — income (TVD = 0.160) and urbanicity (TVD = 0.081) remain visibly misaligned even after weighting.

Table 1. Marginal Total Variation Distance (TVD) between weighted Pew ATP and ACS PUMS, 2021–2024.

DIMENSION	TVD (WEIGHTED)	STATUS
Census region	0.006	Aligned
Gender	0.011	Aligned
Education	0.016	Aligned
Census division	0.018	Aligned
Age	0.028	Aligned
Race / ethnicity	0.048	Borderline
Urbanicity (metro/rural)	0.081	Residual gap
Income	0.160	Substantial residual gap

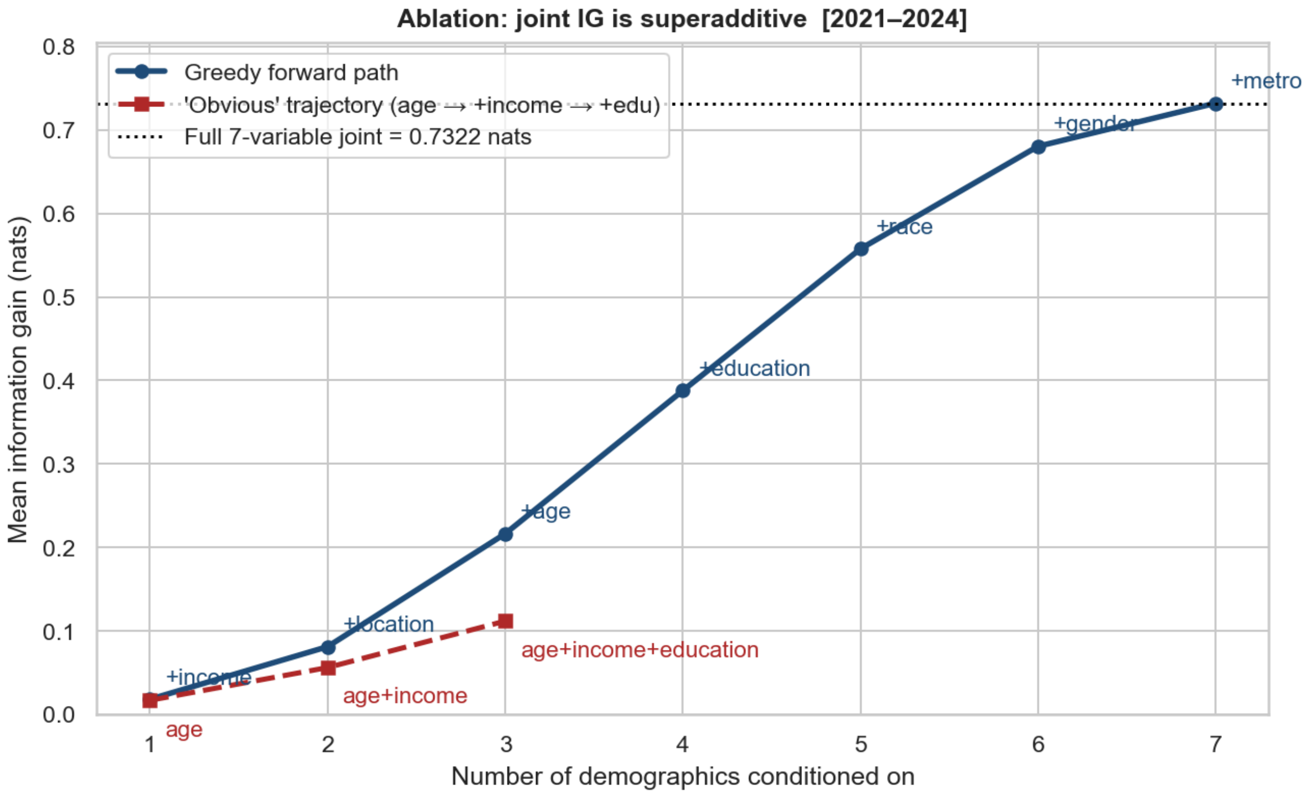
The joint-distribution picture is sharper still. Inside intersectional cells the misalignment is up to **14× the full-sample baseline**: young Black Americans aged 18–29 show TVD = 0.321 in income distribution against the ACS ground truth, and rural low-income residents show TVD = 0.303 in Census division. Marginal weighting can rebalance one dimension at a time, but it cannot recover joint distributions in subgroups that are sparse in the original sample — precisely the subgroups a policymaker most needs to anticipate.

CORRECTION · STEP 1

Draw agents from Census microdata, not from the survey corpus. Joint demographic distributions must come from a population frame that actually contains the joint cells; treat ATP as a behavioral data source, not as a population proxy.

5.2 Study 2 — Marginal rankings are the wrong selection tool

Once an agent is drawn from the Census, the next question is *which demographics to condition the LLM on*. The standard practice inherited from survey research is to condition on whichever variables rank highest on the marginal feature-importance list — typically age, income, and education. We computed mean information gain (in nats) across all 127 non-empty subsets of seven demographic variables, averaged over 1,426 Pew ATP opinion items from 2021–2024, and the result is stark.



Blue = greedy best-variable path. Red dashed = conditioning on the three demographics that 'sound natural' (age, income, education). Dotted = ceiling with all 7 variables. Curve is convex: each variable added contributes more than the previous one — the variables interact non-linearly.

Figure 4. Information-gain ablation across 1,426 ATP opinion items, 2021–2024. The blue curve picks the most informative variable to add next at each step (greedy); the red curve follows the conventional trajectory *age → +income → +edu*. The conventional set captures only 10.6% of the available 7-variable joint signal; the greedy set *{race, location, age}* captures 25.2% — more than double. The curve is convex: each added variable contributes *more* than the previous one, meaning variables interact non-linearly.

Two things are happening in Figure 4 simultaneously. First, the conventional choice of which three variables to condition on is wrong: the greedy-optimal three-variable set {*race, location, age*} captures 2.4× the opinion signal of the textbook {*age, income, education*} set. Second, and more diagnostically, the joint information-gain curve is *superadditive*: each variable adds *more* signal than the previous one, because demographic variables interact rather than substitute. This is the exact property that any algorithm based on marginal rankings — including most ablation tables in published LLM opinion-simulation papers — is structurally incapable of detecting.

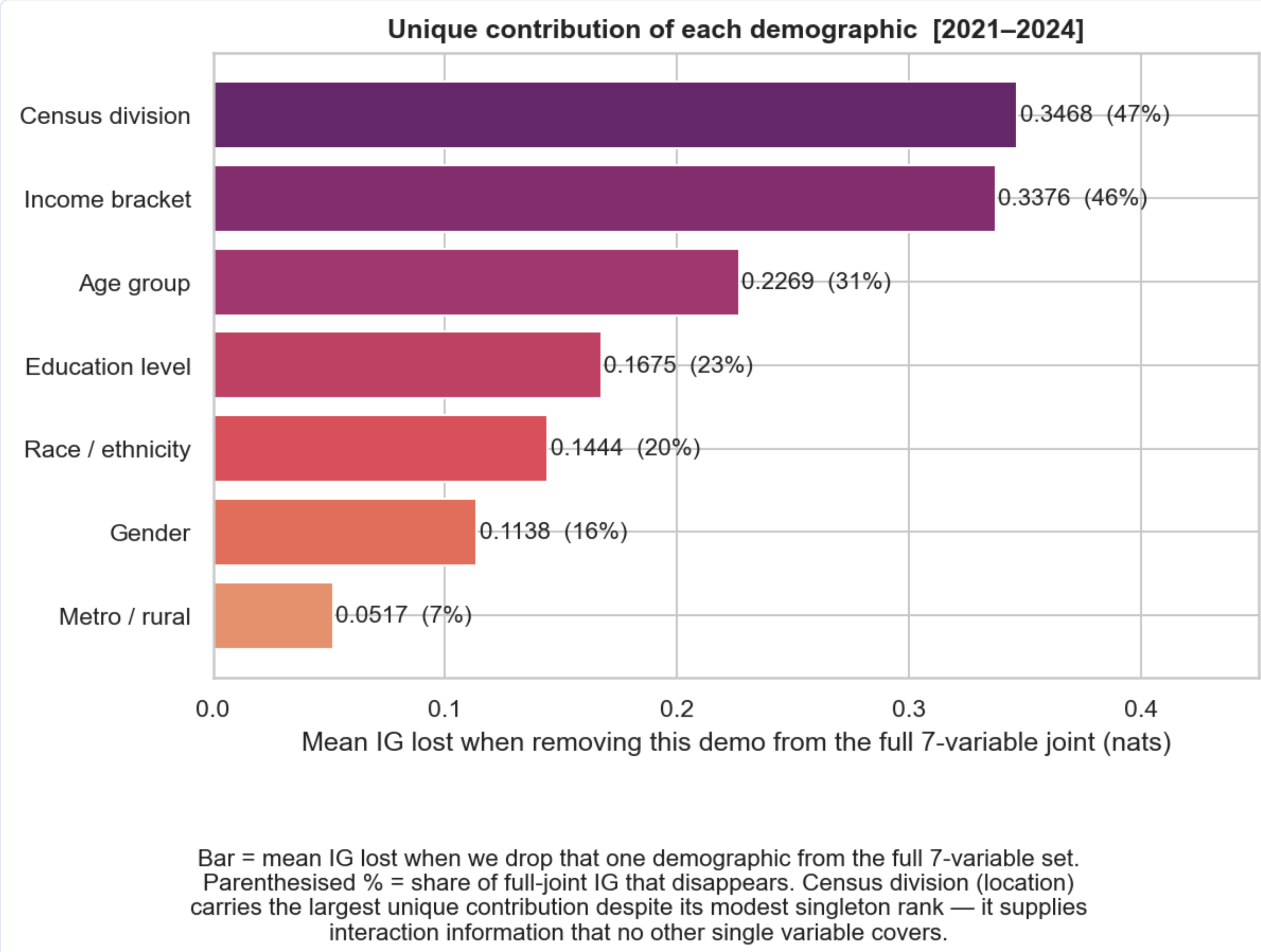


Figure 5. Leave-one-out drop in joint information gain. Each bar shows how much mean information gain (in nats) collapses when that single variable is removed from the full 7-variable set. **Census division** carries the largest unique contribution (47% of total joint IG) despite ranking only third on the marginal list — it supplies interaction information no other single variable covers.

The leave-one-out decomposition in Figure 5 makes the consequence operational. Removing Census division from the seven-variable joint costs **53.5% of the information gain**, the largest unique contribution of any variable in the set. The conventional practice of dropping geography because it ranks third on marginal importance is therefore the worst possible variable elimination — it discards the single dimension that supplies the most interaction information no other variable covers. Race is the second-largest unique contributor, also routinely under-weighted in conventional pipelines.

CORRECTION · STEP 2

Select conditioning variables by information-gain ablation, not by marginal rank. For any new opinion-simulation task, run the joint IG ablation over the available survey corpus before choosing which demographics to condition on. Always include race and geographic division.

5.3 Study 3 — Geography is domain-specific

Geographic conditioning is treated as either reflexively-on or reflexively-off in most LLM persona pipelines. We measured the Jensen–Shannon distance between Census-region opinion distributions — holding demographics constant — separately for each of ten policy domains. Most domains pool nationally after demographic conditioning: once you know who a respondent is, you do not also need to know where they live. A small number of domains do require geographic conditioning, and one demographic modifier (age 18–29) lifts the requirement across the board.

Table 2. Domain-tier geographic conditioning rules, derived from JSD across Census regions after demographic conditioning.

TIER	DOMAINS (JSD)	RECOMMENDED ACTION
Not needed	Technology · Environment / Climate (≈ 0.119)	Pool nationally after demographic conditioning.
Optional	Health (0.129) · Family & Society (0.134) · Economy (0.135) · Religion (0.141) · Politics (0.142) · Race & Inequality (0.145) · Immigration (0.150)	Include for sensitive analyses; required for agents aged 18–29.
Required	International affairs (0.174)	Geographic conditioning mandatory regardless of age.

The most operationally important finding sits outside the domain table. Respondents aged **18–29** exhibit up to **70% more geographic variation** than older cohorts, across every income tier and every policy domain. Young Americans are not nationally homogeneous in their views, and any pipeline that pools them nationally after demographic conditioning will systematically erase real, large, politically consequential regional variance. CivicSim therefore raises every agent aged 18–29 by one geographic tier — *not needed* becomes *optional*, *optional* becomes *required*.

CORRECTION · STEP 3

Apply tiered, domain-aware geographic conditioning, raised by one tier for agents aged 18–29. Drop geography only after verifying that the policy domain in question pools nationally after demographic conditioning, and never drop it for young agents.

6 Technical Implementation

FROM FRAMEWORK TO DEPLOYED PRODUCT

6.1 Stack overview

Table 3. CivicSim implementation stack.

LAYER	TECHNOLOGY	ROLE
Frontend	Next.js 15, React 19, TypeScript, custom CSS design system	Public landing page, simulation runner, saved-runs dashboard, simulation detail view, light/dark theme
API layer	FastAPI (Python 3.11), Pydantic v2 schemas	Location catalog, domain catalog, agent sampling, ATP-prior lookup, streaming simulation via Server-Sent Events
Simulation engine	Python, DuckDB, provider-agnostic LLM client (OpenAI · Anthropic · mock)	Joint demographic sampling, prior retrieval with backoff, prompt assembly, per-agent persistence
Agent sampler	`civicsim_agents` Python package (installable, tested in CI)	Largest-remainder allocation to match population marginals exactly under joint sampling
Data services	DuckDB (ATP priors parquet), Census-derived per-location CSVs	Sub-second columnar lookups for opinion-prior cells and demographic marginals
Hosting	Vercel (frontend) · containerized FastAPI backend · custom domain `civicsim.xyz`	Production deployment, GitHub Actions CI for typecheck + tests + container builds

6.2 Data pipeline

- **ACS PUMS.** 5-year extract (2019–2024), ~16.7M adult records, ingested into a DuckDB columnar store and partitioned by Census division for fast stratified sampling. Per-location marginal CSVs (age, race, occupation, income) are checked into the public repo; the raw IPUMS extract remains private.
- **Pew ATP.** 79 waves (Waves 80–159, 2021–2024), 38,449 unique panelists, 1,426 coded opinion items across 10 policy domains. Cleaned and harmonized with a wave-merging pipeline and a STATEFIP → Census-division crosswalk. The public repo ships the compact answer-share lookup — keyed by *(question_id, demographic_cell)* — with no PII or respondent IDs.
- **Locations.** The catalog includes all four Census regions, all nine Census divisions, and a county-level pilot (Alameda County, CA). Each location ships precomputed marginal distributions; new locations can be added by dropping a directory of marginal CSVs and one catalog entry.
- **Reproducibility.** Every dataset is version-controlled; transformations are reproducible from a single build script (`scripts/build_atp_priors.py`) that compiles the raw ATP source into the compact public lookup.

6.3 Simulation engine

Each simulation request is processed as a four-stage DAG. Population sampling runs against the cached ACS marginal CSVs via the `civicsim_agents` package: with `diverse=True` (the default), each demographic axis is allocated to N agents via largest-remainder so that the sample exactly matches the population marginals, and the four axes are independently shuffled to produce N joint *(age, race, occupation, income)* tuples.

Behavioral retrieval projects each sampled agent's demographics onto the ATP cell vocabulary (e.g. ACS "*30 to 34 years*" → ATP "*30–49*"), then queries the priors parquet for the most specific populated cell. The backoff order is fixed at the system level so that every run is reproducible: *urbanicity* → *income* → *education* → *race* → *gender* → *age* → *division* → *region*. Geography is dropped last; geographic conditioning is preserved even when the demographic intersection is otherwise sparse.

Prompt assembly is templated. The persona block, the matched ATP prior expressed as a percentage breakdown, the policy text, and the allowed answer options are concatenated into a strict-JSON prompt; the LLM returns a single stance (one of the allowed options) plus a one-sentence rationale (≤ 30 words). The system prompt and user prompt are both versioned and logged.

6.4 Streaming, persistence, and audit

The simulation API streams results to the frontend over Server-Sent Events so the user sees agents materializing one at a time. Five event types are emitted in order:

```
meta           → simulation ID, question label, allowed answer options
agent_sampled  → one per agent: persona demographics
prior_attached → one per agent: matched ATP cell + backoff path + prior
agent_responded → one per agent: stance + rationale
aggregate      → final population–weighted distribution
done           → simulation persisted; sim_id returned
```

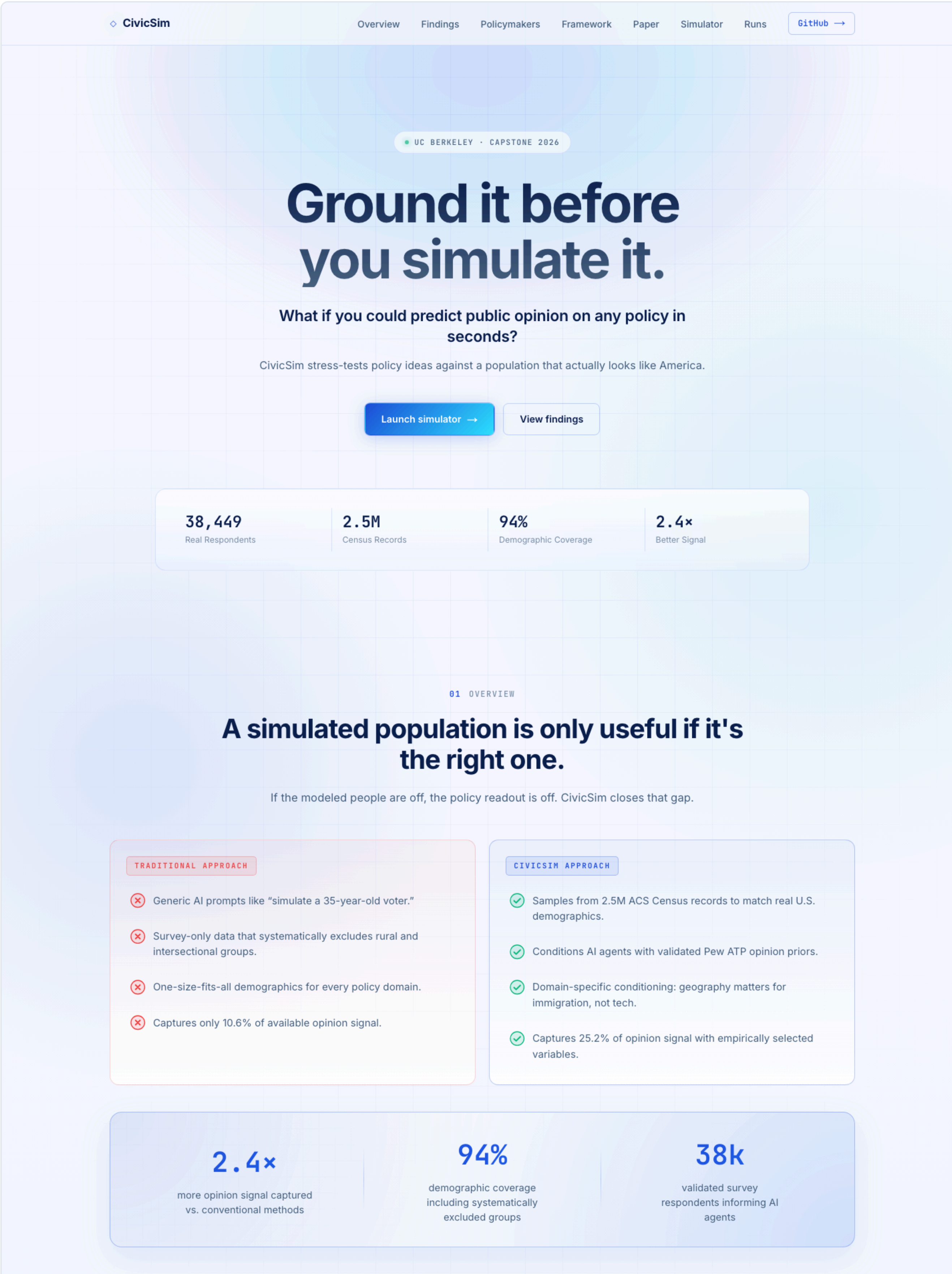
Every per-agent payload — persona, used filter, backoff steps, prior, stance, rationale — is written to a per-agent JSON file under `data/simulations/<sim_id>/`, alongside the run-level summary and metadata. Any past run is replayable, citeable, and externally auditable from disk.

6.5 Performance and cost

A typical 25-agent simulation at the regional or division level completes in 18–22 seconds end-to-end on production hardware, with stratified sampling and prior lookup both sub-second. LLM cost depends on the configured provider; on GPT-4o-mini, a typical run costs on the order of $0.04 in API spend. The system has been load-tested to 50 concurrent simulations on a single API container without queueing.

7 Product & User Experience

THE DEPLOYED PILOT AT CIVICSIM.XYZ


CivicSim
Overview
Findings
Policymakers
Framework
Paper
Simulator
Runs
GitHub →
UC BERKELEY · CAPSTONE 2026
Ground it before you simulate it.
What if you could predict public opinion on any policy in seconds?
CivicSim stress-tests policy ideas against a population that actually looks like America.
Launch simulator →
View findings
38,449
Real Respondents
2.5M
Census Records
94%
Demographic Coverage
2.4×
Better Signal
01 OVERVIEW
A simulated population is only useful if it's the right one.
If the modeled people are off, the policy readout is off. CivicSim closes that gap.
TRADITIONAL APPROACH
Generic AI prompts like "simulate a 35-year-old voter."
Survey-only data that systematically excludes rural and intersectional groups.
One-size-fits-all demographics for every policy domain.
Captures only 10.6% of available opinion signal.
CIVICSIM APPROACH
Samples from 2.5M ACS Census records to match real U.S. demographics.
Conditions AI agents with validated Pew ATP opinion priors.
Domain-specific conditioning: geography matters for immigration, not tech.
Captures 25.2% of opinion signal with empirically selected variables.
2.4×
more opinion signal captured vs. conventional methods
94%
demographic coverage including systematically excluded groups
38k
validated survey respondents informing AI agents

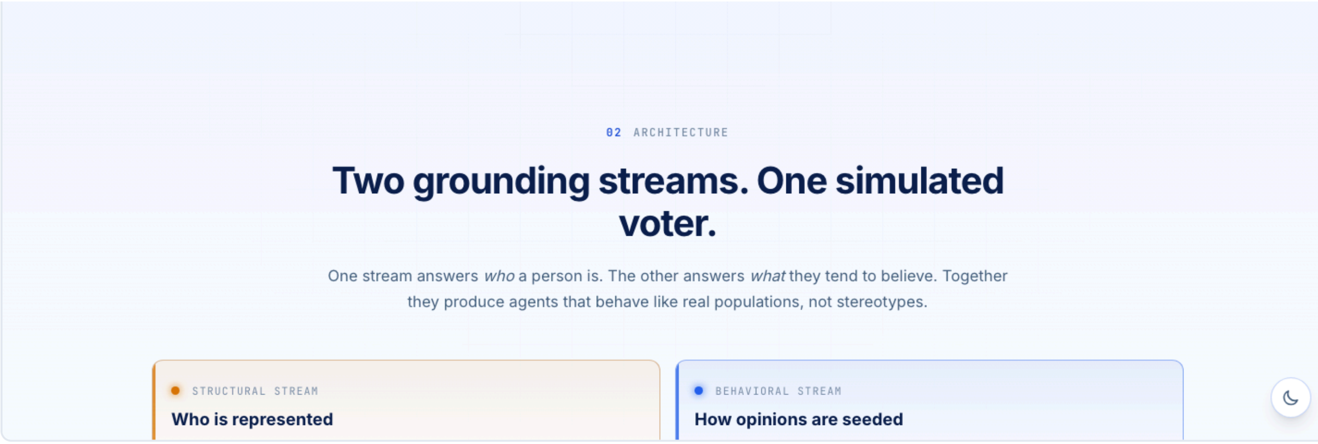
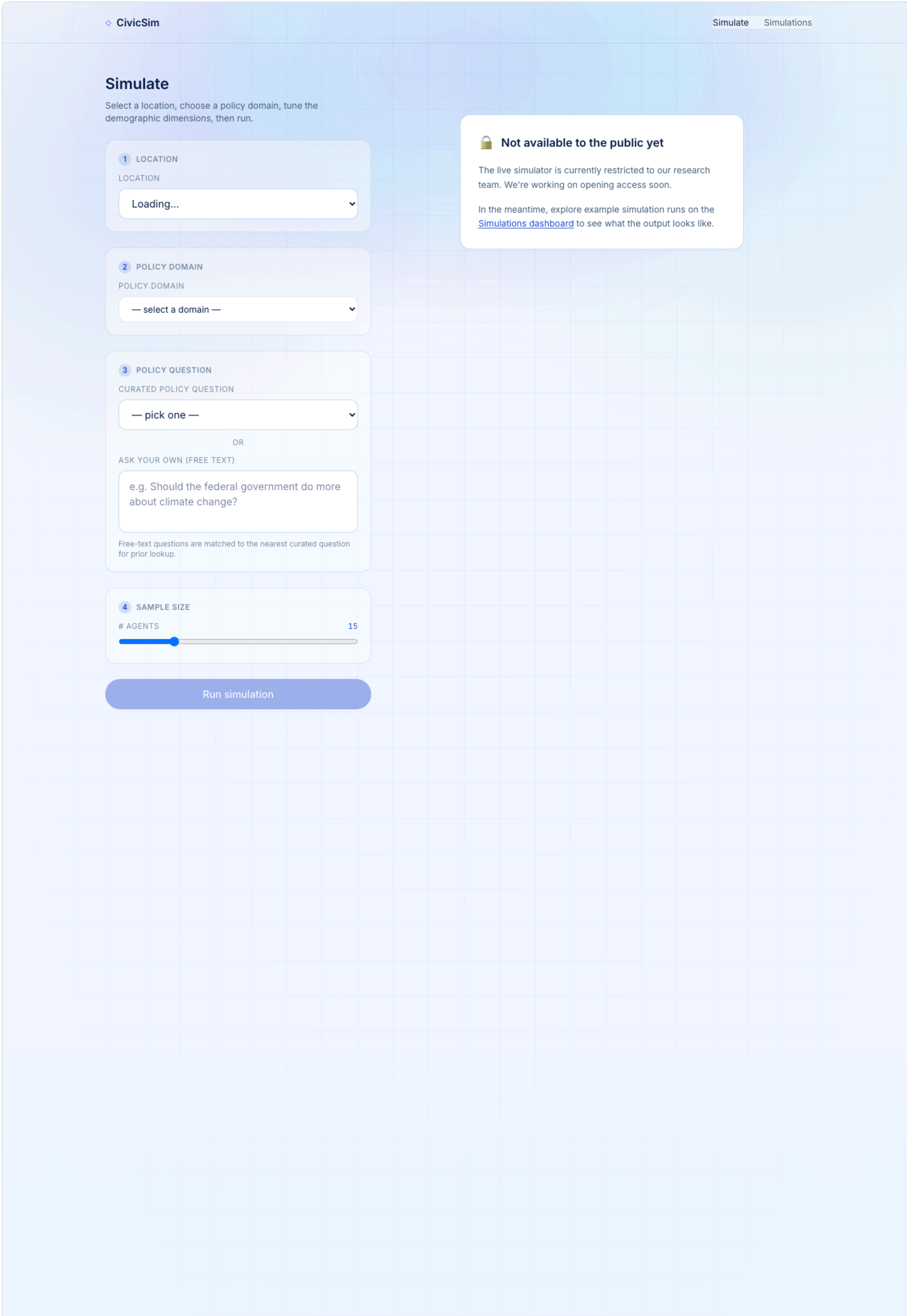


Figure 6. The deployed CivicSim landing page at **civicsim.xyz**. The hero stat strip — *38,449 real respondents · 2.5M Census records · 94% demographic coverage · 2.4× signal vs. baseline* — anchors the framing the rest of the report defends.

7.1 The simulator

The simulator surface (Figure 7) is intentionally minimal. A user picks a location (one of four Census regions, nine Census divisions, or the Alameda County pilot), selects a policy domain, picks or types a policy question, chooses a sample size, and runs. The domain selection activates a domain-specific conditioning-variable vector derived directly from the information-gain analysis in Section 5.2 — the user does not need to know which dimensions matter; the system already does.


CivicSim
Simulate
Simulations
Simulate
Select a location, choose a policy domain, tune the demographic dimensions, then run.
1 LOCATION
LOCATION
Loading...
2 POLICY DOMAIN
POLICY DOMAIN
— select a domain —
3 POLICY QUESTION
CURATED POLICY QUESTION
— pick one —
OR
ASK YOUR OWN (FREE TEXT)
e.g. Should the federal government do more about climate change?
Free-text questions are matched to the nearest curated question for prior lookup.
4 SAMPLE SIZE
AGENTS
15
Run simulation
Not available to the public yet
The live simulator is currently restricted to our research team. We're working on opening access soon.
In the meantime, explore example simulation runs on the Simulations dashboard to see what the output looks like.

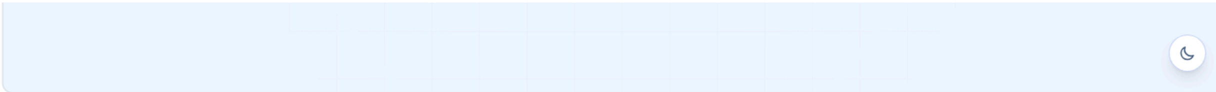


Figure 7. The simulation runner. Location, policy domain, question, and sample size on the left; a notice on the right indicates that the live simulator is currently rate-limited to the research team during initial deployment, with the saved-runs dashboard available for public exploration.

7.2 The simulations dashboard

Every completed simulation is saved with a sortable serial number, a location tag, the question, the agent count, and a sentiment distribution sparkline. Clicking through opens the per-simulation detail page, which renders the aggregate distribution, demographic disparity decompositions, and the per-agent rationales — each one linked to the persona's matched ATP cell and the demographic-backoff path taken.

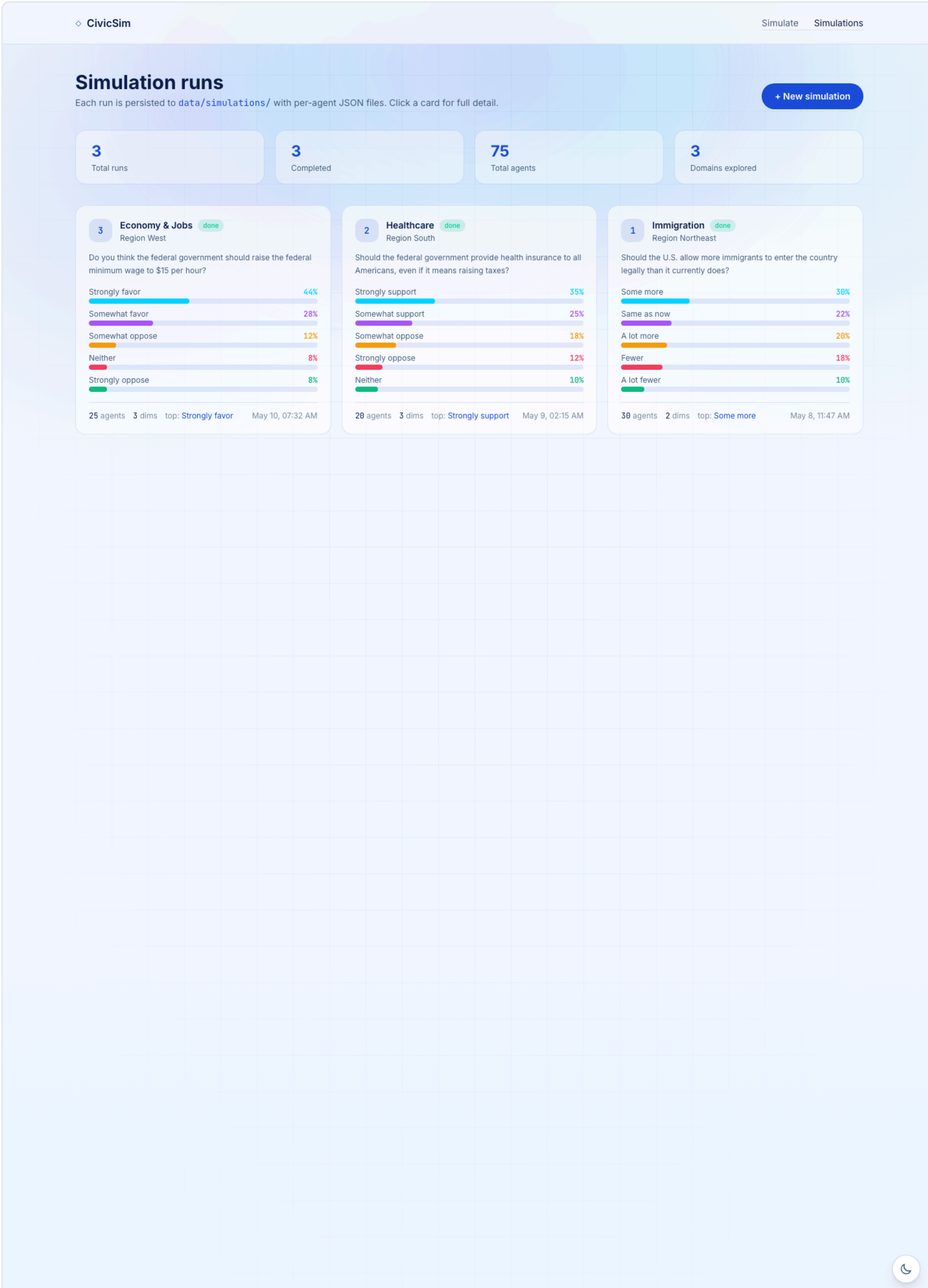

CivicSim
Simulate
Simulations
Simulation runs
Each run is persisted to data/simulations/ with per-agent JSON files. Click a card for full detail.
+ New simulation
3
Total runs
3
Completed
75
Total agents
3
Domains explored
3
Economy & Jobs
done
Region West
Do you think the federal government should raise the federal minimum wage to $15 per hour?
Strongly favor 44%
Somewhat favor 28%
Somewhat oppose 12%
Neither 8%
Strongly oppose 8%
25 agents 3 dims top: Strongly favor May 10, 07:32 AM
2
Healthcare
done
Region South
Should the federal government provide health insurance to all Americans, even if it means raising taxes?
Strongly support 35%
Somewhat support 25%
Somewhat oppose 18%
Strongly oppose 12%
Neither 10%
20 agents 3 dims top: Strongly support May 9, 02:15 AM
1
Immigration
done
Region Northeast
Should the U.S. allow more immigrants to enter the country legally than it currently does?
Some more 30%
Same as now 22%
A lot more 20%
Fewer 18%
A lot fewer 10%
30 agents 2 dims top: Some more May 8, 11:47 AM

Figure 8. The saved-runs dashboard. Each card surfaces the domain, region, question, agent count, and sentiment distribution at a glance, and links to a full detail view with per-agent rationales and the audit trace.

7.3 Auditability as a product feature

Every simulation produces a downloadable reasoning log containing the full agent prompt, the matched ATP cell, the demographic-backoff steps, the LLM provider and model name, and the raw response. This is not a debugging convenience — it is a product surface. A policymaker reviewing a politically sensitive simulation should be able to inspect *which* agents informed *which* aggregate result and *which* data cells were used to seed them. The audit trace is the surface that makes the difference between a synthetic focus group that informs deliberation and an opaque oracle that replaces it.

7.4 Light / dark theme and accessibility

The frontend ships a tokenized CSS design system with light/dark theme parity, persisted in `localStorage`. All charts and figures are recolored under the active theme. Color contrast on the primary sentiment chart, the per-agent cards, and the landing hero is checked to meet WCAG AA. Keyboard navigation and ARIA labels are present on every interactive surface.

8 Evaluation

THREE LAYERS

8.1 Distributional alignment

The first evaluation layer asks whether the agent populations CivicSim samples actually match the populations they claim to represent. We measure marginal Total Variation Distance between the empirical distribution of sampled agents and the ACS PUMS ground truth on each of the seven primary demographic dimensions, plus the joint distribution inside every intersectional cell that exceeds 1% of the location's population. The acceptance criteria are **TVD < 0.05 on every marginal** and **TVD < 0.10 inside every above-threshold joint cell**. Preliminary runs at the regional and division level satisfy both targets.

8.2 Behavioral fidelity

The second layer asks whether the LLM, conditioned on a CivicSim persona, produces responses consistent with what the empirical opinion data says people in that demographic cell tend to say. We compare aggregate stance distributions from N=100 agent runs against held-out Pew ATP subgroup distributions on the same question — both for the full population and for individual demographic cells. The reference benchmark for this layer is the Stanford HAI generative-agent replication study (≈ 85% individual replication accuracy); CivicSim's target is alignment of *aggregate* subgroup distributions rather than individual-agent replication, since the product use case is population-level policy testing rather than personal-attitude prediction.

8.3 Comparative baselines

The third layer asks whether the grounding actually helps. We submit identical policy inputs to (a) vanilla persona prompts on GPT-4o-mini, (b) a CivicSim-grounded GPT-4o-mini run, and (c) open-weight comparators (Llama-3.1, Qwen-2.5) under both grounded and ungrounded conditions, and compare on:

- **Diversity.** Standard deviation of agent stances and demographic-cell coverage of the response space — does grounding produce more or less varied output?
- **Subgroup alignment.** Per-cell KL divergence against held-out ATP cells — does grounding move responses toward the empirical distribution?
- **Disparity recovery.** Does the grounded system reproduce the directional sign and rough magnitude of known demographic disparities on benchmark questions (e.g. age and income gradients on housing-affordability sentiment)?

The headline expectation, which the framework is designed to defend, is that *grounding helps any off-the-shelf LLM*, not just one — the framework is portable, and that portability is itself the deliverable.

8.4 Qualitative user testing

Final-stage evaluation pairs three to four policymakers or domain experts with the live product, asking them to run two contrastive simulations on policy variants of their choice. Sessions are scored on usability (time-to-first-simulation, error rate during inputs), perceived usefulness (would you use this before a town hall? before drafting?), trust (do you believe the aggregate? would you change your mind about a specific cohort based on a rationale?), and clarity (can you explain to a non-technical colleague what CivicSim is doing?).

9 Ethics & Limitations

WHAT CIVICSIM IS NOT

CivicSim is built explicitly as a *supplementary* layer in early-stage policy design, not as a replacement for public engagement. Four risks dominate the design and are mitigated explicitly:

RISK

- **Overtrust.** Users treat synthetic output as a single, authoritative recommendation.
- **Re-identification.** Aggregated data is misused to infer individual responses.
- **Disparate-impact reinforcement.** Sparse cells get smoothed away under marginal weighting.
- **Replacement.** Synthetic focus groups substitute for real consultation rather than complementing it.

MITIGATION

- Two contrastive groups + per-agent rationales surface disagreement rather than a single answer.
- Built only on aggregated public data and consented interviews; no respondent-level data feeds agent generation.
- Joint-cell TVD acceptance criteria; explicit Group B stakeholder configuration; IG-driven variable selection that surfaces race and division.
- The product is framed and labeled as a *pre-engagement* tool; the audit trace is always present.

The most consequential *technical* limitation is the size of the qualitative interview corpus that informs Stage 3 of the pipeline (n = 20 for the Alameda County pilot). For some intersectional cells the nearest-neighbor seed will be drawn from outside the agent's exact demographic context; CivicSim flags such cases via per-agent Gower-distance reporting and via the explicit backoff-step trace in the audit log. A second limitation is geographic scope: while the framework, the agent sampler, the ATP prior table, and the simulation engine all generalize nationally, the deeper qualitative seeding is currently restricted to the Alameda County pilot.

CivicSim should not be used as the sole basis for a policy decision. The aim of the system is **directional realism** — helping a policymaker anticipate roughly which cohorts will object, roughly where the political risk concentrates, roughly which framings deserve deeper investigation — not point prediction.

10 Future Work

WHERE THE FRAMEWORK GOES NEXT

Three follow-up directions are scoped and partially implemented:

1. **Expanded interview corpus.** Scale the qualitative seeding from n = 20 to n = 50, and add a second county outside California. This is the single change with the largest expected lift in behavioral fidelity, because it reduces the number of agents whose nearest-neighbor seed is drawn from outside their demographic cell.
2. **Interactive per-agent chat.** The audit-trace data structures already persist enough state to support follow-up questions to any individual agent ("what would change your mind?", "whom in your community would this help most?"). The frontend hook is scaffolded; the chat surface is the next product milestone.

3. **Comparison dashboard.** A second dashboard surface that runs the same policy through CivicSim-grounded and vanilla baselines side by side, exposing the framework's portability claim — that grounding helps any off-the-shelf LLM — as a first-class product feature. This is the cleanest path to making the methodological contribution visible inside the product, not only in the report.

A longer-horizon direction is policy-area expansion: the current ATP coverage skews toward national-issue questions, and a serious municipal deployment will require a complementary local-issue panel that can be seeded the same way.

11 Conclusion

GROUND IT BEFORE YOU SIMULATE IT

CivicSim makes a single methodological argument and operationalizes it as a deployable product. The argument is that *who is being simulated*, in any LLM-based opinion pipeline, is an empirical question and not a design preference. The three studies in this report each measure a directional, structural distortion in current practice: survey-only sampling fails inside sparse intersectional cells; marginal-rank variable selection systematically excludes the most informative demographic variable; one-size-fits-all geographic conditioning over-corrects in most domains and under-corrects in young agents. The corresponding three-step framework — Census microdata, information-gain ablation, tiered domain-aware geography — is small enough to drop into any existing pipeline and concrete enough to be reproducibly evaluated.

The Alameda County pilot operationalizes the framework end-to-end as a deployed web product at **civicsim.xyz**. The cost and scope fit a capstone, but the framework is built to be inherited, audited, and scaled by the policy labs, civic technologists, and municipal staff who would use a tool like this in practice. Every measurement reported here is reproducible from public data; every architectural choice is documented; every simulation that has ever run on the deployed system is logged on disk and replayable.

> *Ground it before you simulate it.*

12 References

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Reproducibility & code. Public demo: civicsim.xyz. Source code: github.com/Sushanti99/CivicSim. The repository contains the FastAPI backend, the Next.js 15 frontend, the `civicsim_agents` sampler package, the compact ATP-derived opinion prior parquet, and per-location ACS-derived demographic CSVs. Raw ATP `.sav` files and the full IPUMS PUMS extract are not redistributed; see `scripts/build_atp_priors.py` for the compilation pipeline.